

This is sample data provided for CALIBER purposes only		ONE POINT LESSON (OPL)		OPL No: OPL-FA-8901-07 Sheet 1 of 1
OPL Title	Manway Box-Up After Internal Inspection	Discipline	Mechanical	
Equipment	FA-8901 - REFLUX ACCUMULATOR DRUM	Area / Unit	8900 - HEXANE COLUMN OVERHEAD SYSTEM	
Related Interlock	SEQ-8901 (SIL 1)	P&ID; Ref	TJC-LLD-PID-8901	
Classification:	☐ Basic Knowledge	☒ Improvement	☐ Trouble Case	Aspect: P-Q-C-D-S-M-E

1. PURPOSE / OBJECTIVE

This One Point Lesson explains **manway box-up after internal inspection** for FA-8901 (REFLUX ACCUMULATOR DRUM) in the HEXANE COLUMN OVERHEAD SYSTEM. Correct gasket selection and bolt torque after inspection prevent a hexane leak from the manway. The lesson gives operators and technicians the key knowledge and the correct step-by-step method so the task is done safely, correctly and consistently every time.

2. SAFETY PRECAUTIONS

Hazard note: this equipment is **HIGH CRITICAL** handling **Hexane / water** at up to 10 barg / FV / 120 degC. Observe the following before and during the task:

- Apply LOTO (Lock-Out Tag-Out) and prove zero energy before starting work.
- Isolate, depressurise, drain and purge the equipment; obtain a valid work permit.
- Wear the correct PPE: coverall, safety shoes, gloves, goggles and hearing protection.
- Beware of stored mechanical energy (springs, rotating inertia) and pinch points.
- Use certified lifting gear and correct manual-handling technique for heavy parts.

3. TOOLS & MATERIALS REQUIRED

- Standard mechanical hand tools & calibrated torque wrench
- LOTO padlocks and permit
- Laser/dial alignment kit (as applicable)
- Feeler gauges, dial indicator, seal/bearing pullers
- Correct spare parts and gaskets per the BOM
- Lifting gear and slings (certified)

4. DETAILED PROCEDURE / STEPS

Step	Action	Check / Acceptance
1	Fit a new SW SS316/graphite manway gasket	Isolation & LOTO verified
2	Torque the manway bolts in a star pattern in passes	Parts match BOM / spec
3	Leak test with the nitrogen blanket	Torque / clearance to spec
4	Re-instate the PSV and all instruments	No leak / smooth operation
5	Confirm the drum is ready before re-introducing hydrocarbon	Record in equipment history

5. COMMON PROBLEMS & TROUBLESHOOTING

Symptom	Likely Cause	Action
Statutory relief valve test	Set pressure drift found	Overhauled PSV-8901, lapped seat, re-set 10 barg, certified
Boot drain valve N4 passing / leaking	Seat eroded by water/solids	Reseated / replaced boot drain valve HV-8904
Scheduled corrosion monitoring of boot	Wall thickness within corrosion allowance	UT survey of boot and shell bottom, logged in RBI file

6. KEY LEARNING POINTS

- Correct gasket selection and bolt torque after inspection prevent a hexane leak from the manway.
- Respect the interlock SEQ-8901: e.g. LSHH-8901 trips at > 85 % - never defeat it without a permit.
- Always cross-check the Datasheet limits and the P&ID; TJC-LLD-PID-8901 before working on FA-8901.
- Record the work in the equipment history so the Knowledge Hub stays the single source of truth.

Prepared by	Reviewed by (Supervisor)	Approved by (Manager)	Date of Sharing
Panel Operator / Technician	Yudha Permana (EMP-1124)	Zulkarnain Hasan (EMP-0901)	Wednesday, 01 July 2026